

Operations on graphs



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- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Operations on graphs

- Knowing the graph of a function $y = f(x)$, by means of simple geometric transformations it is possible to draw the graph of the following functions:

$$y_1 = f(x) + a, \quad a \in \mathbb{R} \quad (1a)$$

$$y_2 = f(x + a), \quad a \in \mathbb{R} \quad (1b)$$

$$y_3 = k f(x), \quad k \in \mathbb{R} \quad (1c)$$

$$y_4 = f(kx), \quad k \in \mathbb{R} \quad (1d)$$

$$y_5 = |f(x)| \quad (1e)$$

$$y_6 = f(|x|) \quad (1f)$$

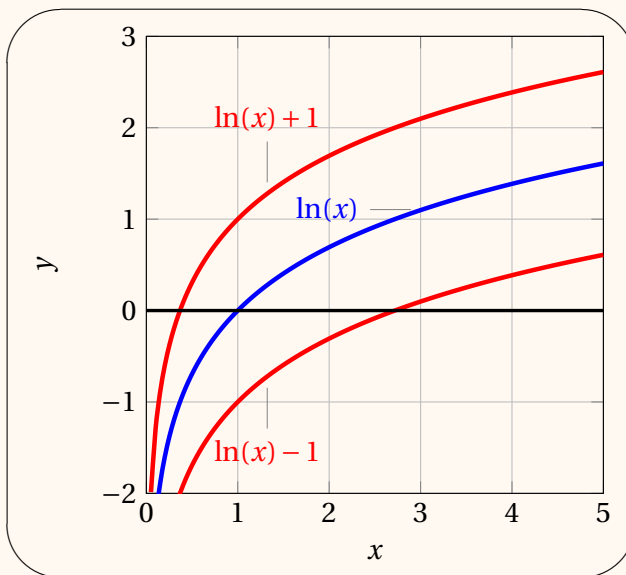
- Therefore, starting from the elementary functions, it is possible to build, by means of these operations, a large variety of new functions. These operations create **translations**, **dilations** and **reflections** of the original function.

1.1 Operations related to $y_1 = f(x) + a$

Example 1: Operations on graphs related to $y_1 = f(x) + a$

Consider $y = \ln x$. Then

$$y_1 = \ln(x) + a \quad (x > 0).$$



- The graph of y_1 is obtained from that of y by a **translation** of a units **upward** if $a > 0$, **downward** if $a < 0$.

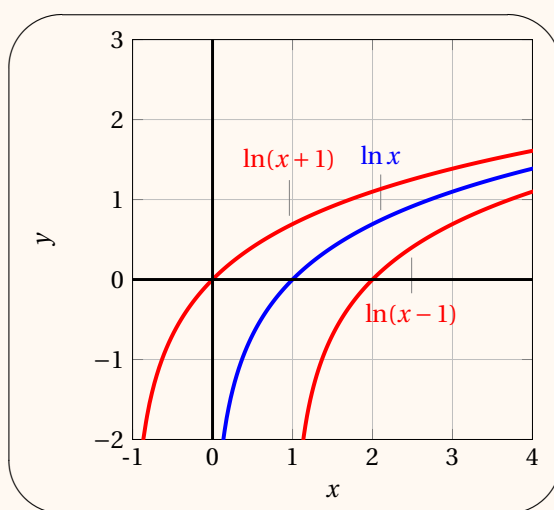
1.2 Operations related to $y_2 = f(x + a)$

Example 2: Operations on graphs related to $y_2 = f(x + a)$

Consider again $y = \ln x$. Then

$$y_2 = \ln(x + a).$$

Since $\ln x$ is defined for $x > 0$, $\ln(x + a)$ will be defined for $x + a > 0$, i.e., for $x > -a$. Since $\ln x = 0$ if $x = 1$, we have $\ln(x + a) = 0$ for $x + a = 1$, i.e., $x = 1 - a$.



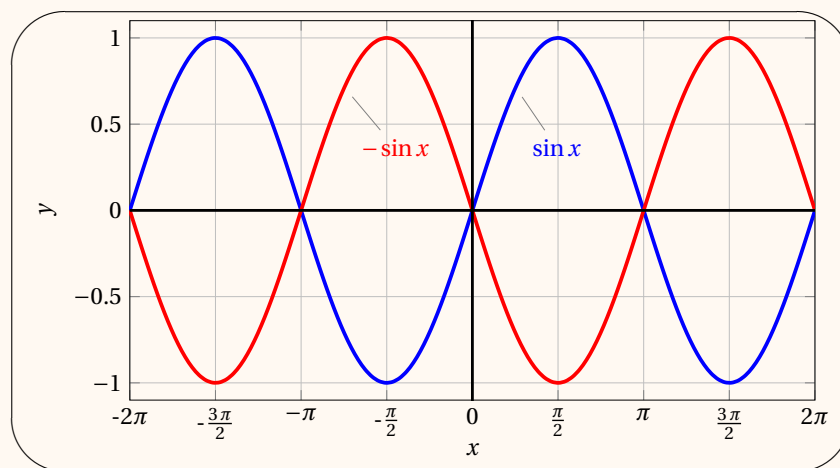
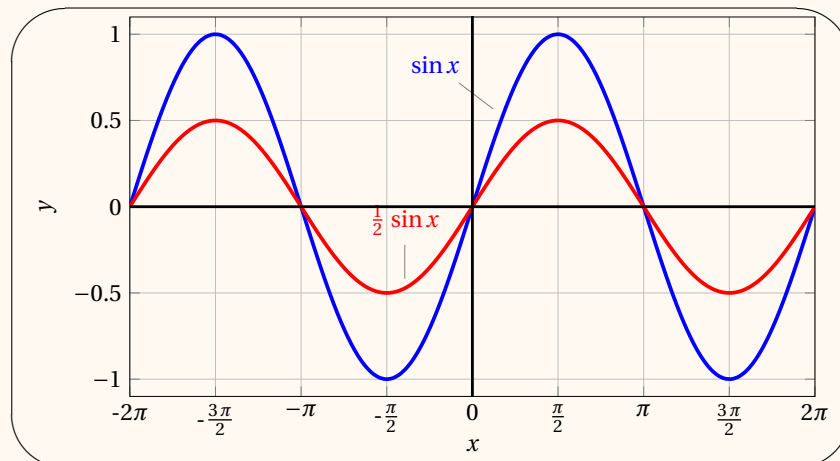
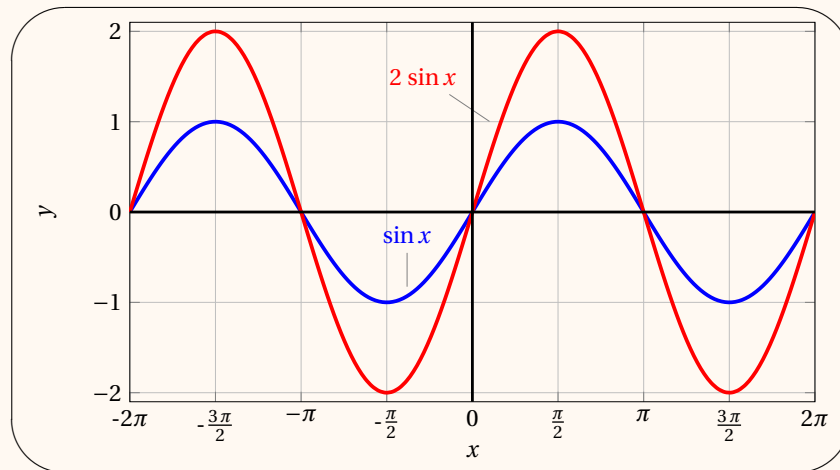
- The graph of y_2 is obtained from that of y by a **translation** of a units to the **left** if $a > 0$, to the **right** if $a < 0$.

1.3 Operations related to $y_3 = k f(x)$

- The graph of $y_3 = k f(x)$ is obtained from that of f by multiplying all the ordinates $f(x)$ by k .
- In particular, if $k = -1$ the ordinates simply change sign, so that the graph of y_3 is symmetric to that of f with respect to the x -axis.
- Note that if $k > 1$, the graph is “**stretched**” in the **vertical direction**, dilating the **positive** ordinates **upward** and the **negative** ones **downward**.
- Conversely, if $0 < k < 1$ the graph “**shrinks**”, again in the vertical direction.
- Therefore, the operation of multiplying $f(x)$ by k has the geometric meaning of a **dilation** (if $|k| > 1$) or a **contraction** (if $|k| < 1$) along the y -axis, possibly accompanied by a **reflection with respect to the x -axis**, if $k < 0$.

Example 3: Operations on graphs related to $y_3 = k f(x)$

Consider $y = \sin x$, then $y_3 = k \sin(x)$.

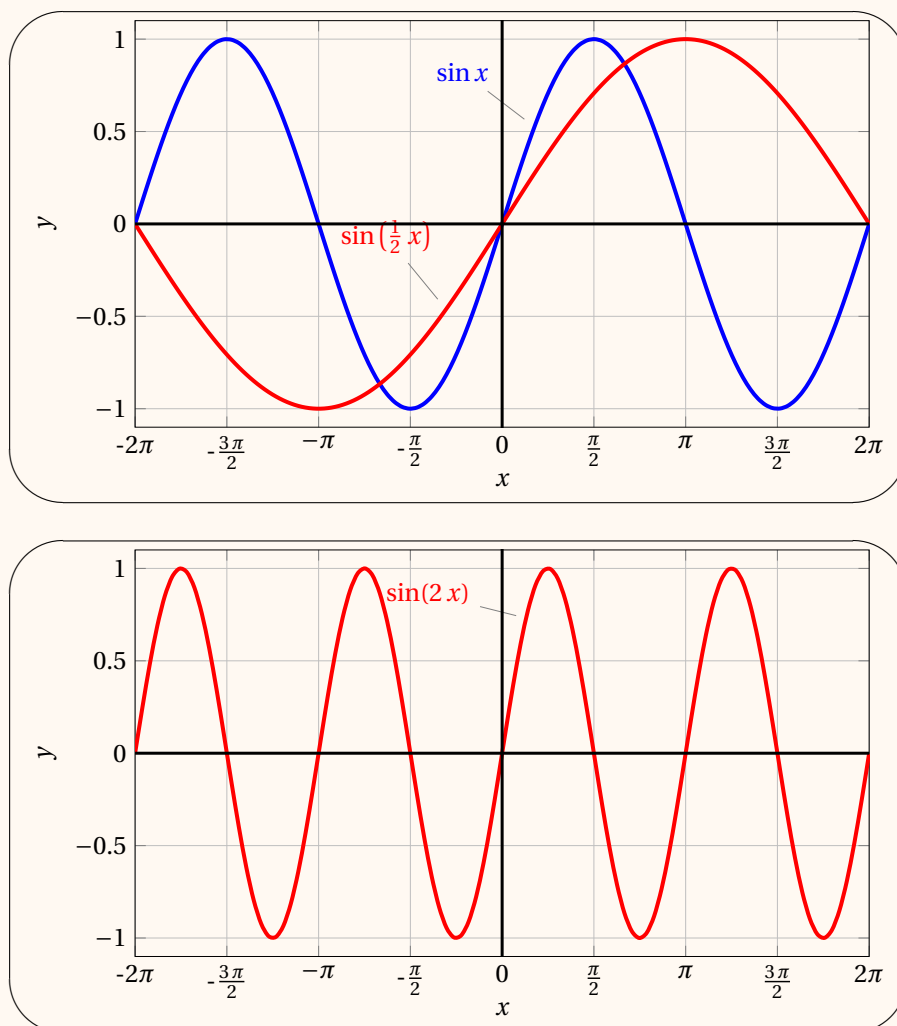


1.4 Operations related to $y_4 = f(kx)$

- The graph of $y_4 = f(kx)$ is obtained from that of $f(x)$ by a **change of scale** along the x -axis.
- If $k > 1$, kx grows faster than x , and therefore the graph of y_4 will be similar to that of f but with faster oscillations, that is, it will be “**compressed**” in the **horizontal direction** by a factor $\frac{1}{k}$.
- Similarly, if $0 < k < 1$ the graph will appear “**dilated**” in the **horizontal direction**, with gentler oscillations.
- If $k < 0$, in addition to a compression (if $|k| > 1$) or a dilation (if $|k| < 1$) along the x -axis, there will be a **reflection with respect to the y -axis**.

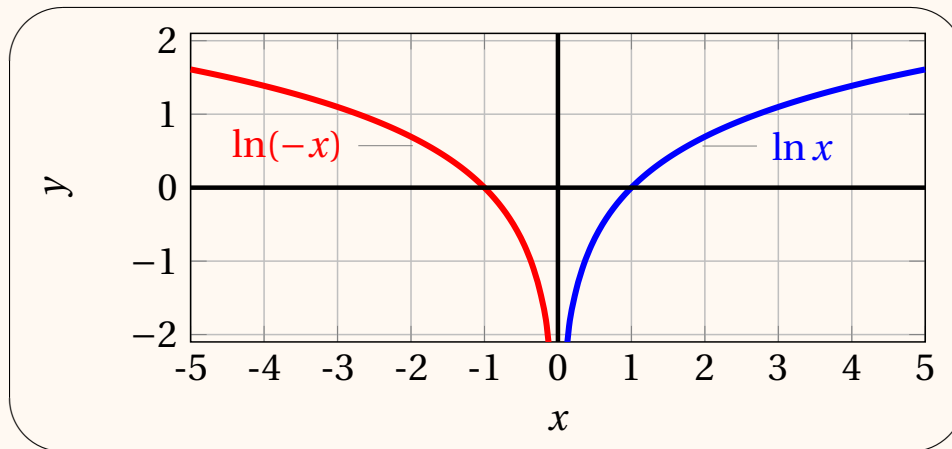
Example 4: Operations on graphs related to $y_4 = f(kx)$

Consider $y = \sin x$, then $y_4 = \sin(kx)$. With $k = \frac{1}{2}$ and $k = 2$ we have:



Example 5: Operations on graphs related to $y_4 = f(kx)$

Consider $y = \ln x$, then $y_4 = \ln(kx)$. With $k = -1$ we have:



1.5 Operations related to $y_5 = |f(x)|$

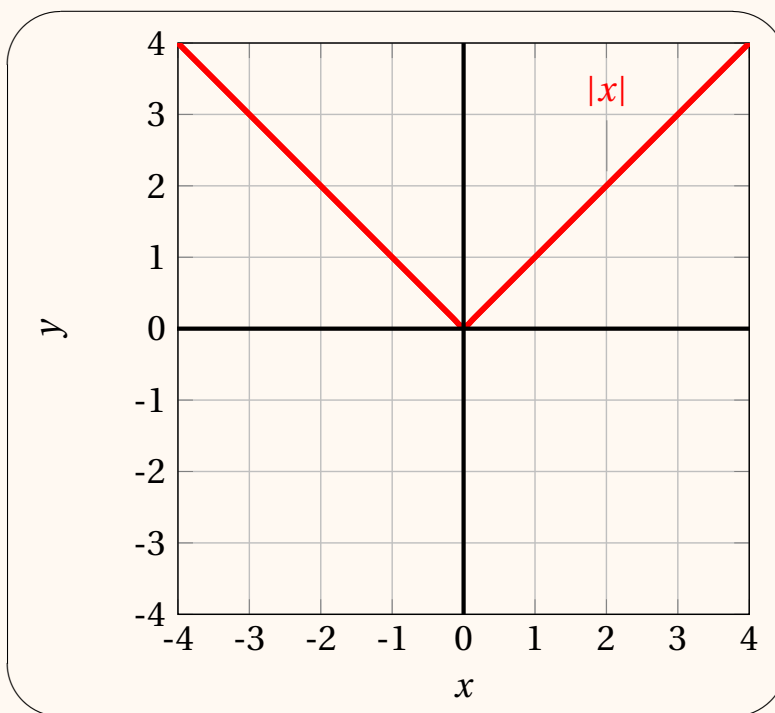
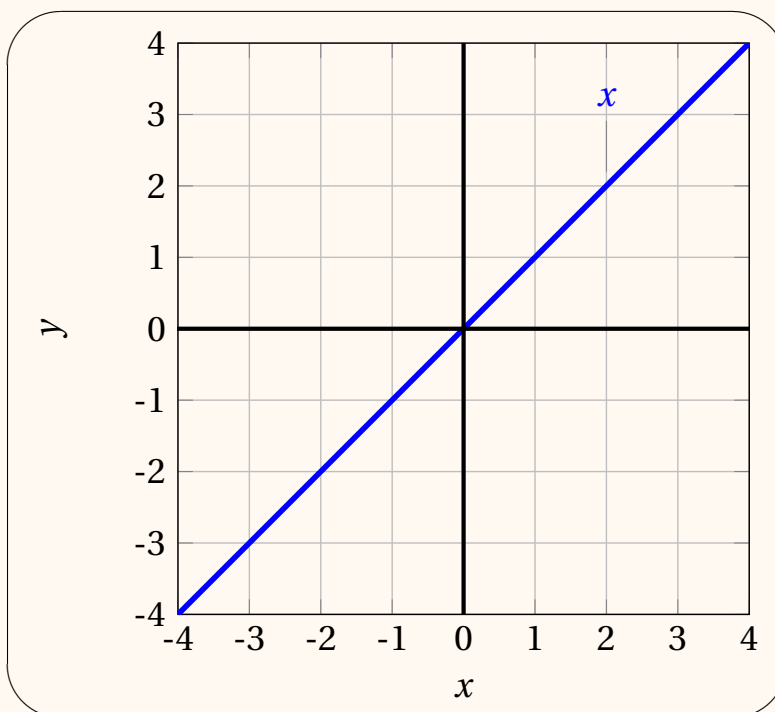
Absolute value of $f(x)$:

$$|f(x)| = \begin{cases} f(x) & \text{if } f(x) \geq 0, \\ -f(x) & \text{if } f(x) < 0 \end{cases}$$

- When passing from the graph of $y = f(x)$ to that of $y_5 = |f(x)|$, the points with **non-negative ordinate remain unchanged**, while those with **negative ordinate are transformed into their symmetric points with respect to the x -axis**.
- The graph of $y_5 = |f(x)|$ is obtained from that of f by “flipping” symmetrically with respect to the x -axis the part of the graph of f that lies in the lower half-plane, and leaving the rest unchanged.

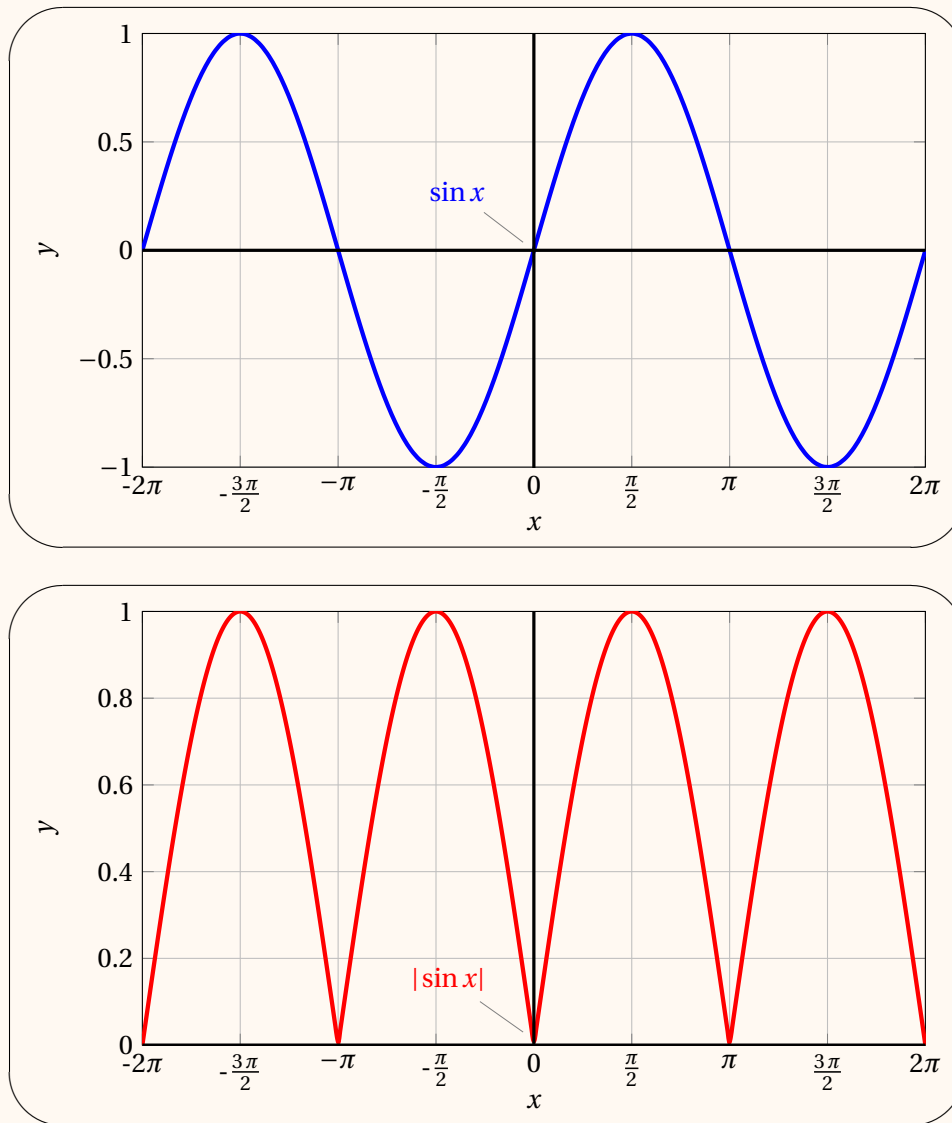
Example 6: Operations on graphs related to $y_5 = |f(x)|$

Consider $y = x$, then $y_5 = |x|$



Example 7: Operations on graphs related to $y_5 = |f(x)|$

Consider $y = \sin x$, then $y_5 = |\sin x|$



1.6 Operations related to $y_6 = f(|x|)$

- Finally, to draw the graph of $y_6 = f(|x|)$, we observe that $|x| = x$ for $x \geq 0$, and hence in the right half-plane the two graphs coincide.
- We have $|-x| = |x|$, and hence y_6 is an even function, therefore symmetric with respect to the y-axis.
- Consequently, the graph of $y_6 = f(|x|)$ is drawn by **leaving the graph of f unchanged in the right half-plane and flipping it symmetrically with respect to the y-axis.**

Example 8: Operations on graphs related to $y_6 = f(|x|)$

Consider $y = e^x$, then $y_6 = e^{|x|}$

